

Hospital Management System

Database Design — Entities, Attributes & Relationships

1. Entities and Attributes

1. Patient

PatientID | Name | Gender | Status | Phone | Email | ReasonOfVisit | Address | AppointmentID

2. Doctor

DoctorID | Name | Gender | Specialization | Qualification | Phone | Email | Address |
DepartmentID (FK) | Experience | JoiningDate | Shift | Status | MaxPatientsPerDay

3. Nurse

NurseID | Name | Gender | Phone | Email | DepartmentID (FK) | Shift | Qualification |
JoiningDate | Status | WardAssigned

4. Department

DepartmentID | DepartmentName | Location | TotalBeds | Floor | Status | Description

5. Appointment

AppointmentID | PatientID (FK) | DoctorID (FK) | AppointmentDate | AppointmentTime | Status |
Reason | RoomNumber

6. Medical Record

RecordID | PatientID (FK) | DoctorID (FK) | Symptoms | VisitDate | NextAppointmentDate |
PrescribedMedicines | Diagnosis (Disease) | Precautions

7. Room / Ward

RoomID | RoomType | FloorNumber | DepartmentID (FK) | DepartmentName | AvailabilityStatus
| NumberOfRooms

8. Billing

BillID | PatientID (FK) | DoctorFee | MedicineFee | PaymentMethod | PaymentDate

9. Staff

StaffID | Name | Gender | Phone | Email | Address | DepartmentID (FK) | JoiningDate | Salary
| ShiftTiming | Status

2. Entity Relationships

Patient

Related Entity	Relationship	Description
Appointment	One-to-Many	a patient can book several appointments
Doctor	Many-to-Many	a patient can consult different doctors for different conditions
Medical Record	One-to-Many	a patient builds up multiple medical records over time

Doctor

Related Entity	Relationship	Description
Department	Many-to-One	a doctor belongs to a single department
Appointment	One-to-Many	a doctor handles multiple appointments
Medical Record	One-to-Many	a doctor creates several medical records
Room / Ward	Many-to-Many	a doctor may attend patients across different wards

Nurse

Related Entity	Relationship	Description
Department	Many-to-One	a nurse is assigned to a single department
Room / Ward	Many-to-Many	a nurse can be assigned to multiple wards
Patient	Many-to-Many	a nurse cares for multiple patients
Doctor	Many-to-Many	a nurse may assist multiple doctors

Department

Related Entity	Relationship	Description
Doctor	One-to-Many	a department employs multiple doctors
Nurse	One-to-Many	a department employs multiple nurses
Room / Ward	One-to-Many	a department contains multiple rooms/wards
Staff	One-to-Many	a department employs multiple staff members

Appointment

Related Entity	Relationship	Description
Patient	Many-to-One	many appointments can belong to one patient

Doctor	Many-to-One	many appointments can belong to one doctor
Room / Ward	Many-to-One	many appointments take place in one room over time

Medical Record

Related Entity	Relationship	Description
Patient	Many-to-One	many records belong to one patient
Doctor	Many-to-One	many records are created by one doctor
Appointment	One-to-One	each record corresponds to a single visit

Room / Ward

Related Entity	Relationship	Description
Patient	One-to-Many	a ward accommodates multiple patients

Billing

Related Entity	Relationship	Description
Patient	One-to-Many	a patient may generate multiple bills over time
Doctor	Many-to-One	many bills can be linked to one doctor
Medical Record	One-to-One	each bill corresponds to one visit record

Staff

Related Entity	Relationship	Description
Department	Many-to-One	a staff member is assigned to a single department
Room / Ward	Many-to-Many	a staff member may be assigned to multiple wards/rooms